

Derek Chen

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EDUCATION

The Cooper Union for the Advancement of Science and Art

Bachelor of Science in Computer Science, GPA: 3.816

New York, NY

Aug. 2025 – Current

EXPERIENCE

Fellow

Spring 2026

CodePath

Remote

- Selected for CodePath's Intermediate Web Development (WEB102) fellowship, an industry-mentored program covering full-stack fundamentals across 10 weekly graded labs and 8 projects
- Shipped ShipHub, a platform for people to share and suggest project ideas, applying React/Supabase and REST API integration under weekly technical assessments reviewed by industry mentors
- Solved 120+ data structures & algorithms problems in Python through CodePath's TIP102 curriculum, completing weekly timed practice sessions to build interview readiness
- Collaborated with industry mentors and a team of 4 peers to code-review and ship features on a fixed project timeline

Fellow

Summer 2025

All Star Code

Remote

- Earned a competitive, application-only fellowship seat by demonstrating fundamentals in computer science, entrepreneurship, and professional development across 25 labs and a final project
- Shipped Affirmation Color, a platform for people to create and share drawings under motivational quotes, building full-stack features with HTML, CSS, JavaScript, and Firebase from design through deployment alongside a team of 3
- Integrated third-party APIs and managed Firebase databases to support real-time data storage and retrieval across the application

PROJECTS

PengPlayer | *C++, Spotify Web API, librespot, MSVC, vcpkg*

2026

- Built a native Windows terminal-based (TUI) Spotify player in C++, implementing OAuth Authorization Code Flow with automatic token persistence and refresh across sessions
- Integrated librespot for gapless audio playback by managing the subprocess via the Windows API and polling for device readiness before playback
- Designed a REST client layer with cpp-httplib and nlohmann/json wrapping 9 Spotify Web API endpoints, supporting full playback control and queue management

ShipHub | *React, JavaScript, Supabase, Netlify*

2026

- Enforced per-user data access on a Postgres-backed discussion board by designing Supabase row-level security policies across posts, comments, and votes, closing off unauthorized reads/writes at the database layer
- Modeled per-post voting with a post_votes join table under a (post_id, user_id) unique constraint, using upsert-based casting so each user holds exactly one active up/down/neutral vote per post
- Optimistically updated vote counts and UI state on click with rollback to the prior value on write failure, then built file uploads and an inline kebab-menu delete flow

Shareboard | *React, TypeScript*

2026 – Present

- Rendered a full-viewport, hand-drawn-style canvas with pointer-capture-based panning, devicePixelRatio-aware resizing, and 10 discrete zoom levels stepped via a throttled (100ms) wheel handler, using rough.js for shape rendering and perfect-freehand for stroke input
- Kept panning smooth under continuous pointer movement by storing viewport offset in a ref instead of React state, driving canvas redraws imperatively and reserving useState only for the far-less-frequent zoom-level changes surfaced to the UI
- Refactored the app from a single-file prototype into a typed, convention-driven file structure, isolating viewport/rendering logic to keep future feature work (hit-testing, multi-tool support) additive rather than requiring further restructuring

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React, P5.js

Databases & Backend: FastAPI, Supabase, PostgreSQL

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Platforms & Tools: Netlify, REST APIs, Git/GitHub